Choose Your Route

Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.

Student name :	Date : .....

PRACTICE — Secure the method

Match it, solve it, check it

Items 1–3: solve, and write which method you used. Items 4–6 are below.

1.	2.	3.
$2^{x-3} = 16$	$27^{x} = 9$	$\log_2(x-1) = 4$

4. Solve $3^{x+1} > 81$. Say whether the direction of the inequality changes, and why.
5. Solve $\log_5(x+4) = 2$ and check that the argument is positive.
6. Solve $6^{x} = 50$: give the exact answer as a logarithm, then to 2 decimal places.

Done when: both sides share one base before the exponents are equated, and each logarithm answer is checked in the original equation.

APPLY — Use it in context

Flip it, reject it, model it

1. Solve $\left(\frac{1}{4}\right)^{x} \leq \frac{1}{64}$. Test one value inside and one outside your answer.
2. Solve $\log_2(x+2) + \log_2(x-1) = 2$. State the domain, and name any rejected value.
3. A model (model values only): the number of subscribers to a Jeddah fibre-internet service is $S(t) = 2000\,e^{0.3t}$ after t years. Find when S reaches 10 000: exact form first, then to 1 decimal place.

Done when: the direction of the inequality is justified by the size of the base, and each logarithm solution is tested against the domain.

INVESTIGATE — Find out why

Arguments, not answers

1. A student solves $\log_3(x-4) + \log_3(x+4) = 2$ and writes $x = 5$ or $x = -5$. Which answers are valid? Explain the error.
2. Explain why $2^x < 8$ gives $x < 3$ but $0.5^x < 8$ gives $x > -3$.
3. Solve $2^x = 3^{x-1}$ exactly. Explain why matching bases cannot work here, and give the answer to 2 decimal places.

Done when: each explanation uses the domain of the logarithm or the size of the base, and the exact answer is given before the decimal.

Remember

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.